1 Introduction

Assume that in an environment, an agent must choose an action from a set of available actions. The utility of this action for the agent can be expressed as a real-valued function and depends on the unknown state of the environment. More precisely, if we denote the action as a and the state of the environment as x, the utility of each action for the agent is a real-valued function $U(a, x)$. We further assume that an evidence e of the environment state exists, and our agent can leverage this evidence to determine the optimal action.

According to Bayesian decision theory, the optimal action, when evidence e is observed, is the one that maximizes the expected utility conditioned on e:

$$a_{opt} = \arg\max_a \mathbb{E}_{P^*}\left[U(a, x) \mid e\right] = \arg\max_a \sum_x U(a, x)P^*(x \mid e).$$

Here $P^*(X \mid e)$ is the probability distribution of the environment state given the evidence. But how can we get this distribution? Is there an objective distribution inherent in the environment that we need to somehow discover?

From the subjective perspective on probability, the answer is negative. There is no such objective distribution, and P^* exists only in our minds. It is a tool that helps us regulate our reasoning, and an intermediate measure which is created inside the Bayesian decision-making framework to ensure we stay consistent with our assumptions and common sense principles.

Thus, if we accept the subjective viewpoint, P^* needs to be determined by one's prior knowledge and assumptions. When we do not have any assumptions and prior knowledge, the set of candidate distributions for P^* is simply the set of all probability distributions. As we make assumptions or use our prior knowledge, the probability distributions that are not consistent with our assumptions are removed from this set, and the set of acceptable probability distributions gets smaller and smaller. We shall call this set a probabilistic model:

Definition 1 (Probabilistic Model)**.** We define a *probabilistic model* $\mathcal{M}$ as a pair $(\mathcal{S}, \mathcal{P})$, where $\mathcal{S}$ is a sample space, and $\mathcal{P}$ is a set of acceptable probability measures on $\mathcal{S}$.

One of the most commonly used assumptions in the statistical modeling is the exchangeability assumption. We say a sequence of random variables $X_1, X_2, \cdots$ is exchangeable if the ordering of variables does not provide any useful information. In other words, for an exchangeable sequence of random variables, the joint probability is the same for any permutation of variables, and this property holds for every subset of the sequence. For example, if each X_i is a binary variable, the joint probability depends only on the total number of 1s and 0s, and the exact ordering of them is irrelevant.

When a sequence of random variables is exchangeable, de Finetti's theorem states that their joint probability has a special representation:

$$P(x_1, x_2, \cdots) = \int \prod_i \pi(x_i \mid \theta) p(\theta) d\theta.$$

This means exchangeable observations are conditionally independent relative to some latent parameter θ.

Moreover, the theorem states that if each X_i is discrete, $\pi(x_i \mid \theta)$ would be the multinomial distribution. So in our decision-making problem, if the evidence e is a sequence of previous experiences of the environment $X_1, X_2, \cdots$, and if we assume that our environment does not change over time so that this sequence is exchangeable, then to fully determine P^*, we only need to determine the prior probability of our latent parameter $p(\theta)$.

Definition 2 (Parametric Model)**.** Let Θ be a random vector. A parametric model is a probabilistic model $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$, where for every $P \in \mathcal{P}$ there is a prior probability distribution p, such that for every $s \in \mathcal{S}$

$$P(s) = \int \pi(s \mid \theta) p(\theta) d\theta, \tag{1}$$

while π is a known probability distribution.

Clearly, one way to select a suitable prior distribution $p(\theta)$ is to use some prior knowledge about the environment. However, what if such knowledge is unavailable?

It is not difficult to identify some distributions that are not good choices for $p(\theta)$. For instance, any distribution that gives zero probabilities to certain values of θ is not a good choice, because if we use such a prior, no evidence can convince us that the value of θ lies within those excluded values. Unfortunately, arguments like this one can only rule out a small portion of the possible distributions for $p(\theta)$, and we are left with many options while it remains unclear which one we should select.

In this discussion, we will explore a new approach to selecting a suitable prior distribution when we lack prior knowledge about the environment.

2 Central Distribution

Assume that in an environment we are interested in the probability of a random variable X conditioned on an observed evidence e. We have decided on a probabilistic model $\mathcal{M}$ for the environment, and without any particular prior knowledge, we have chosen a distribution P from our model. We ask an expert, who is familiar with the environment, to evaluate our distribution P. Presumably, that expert has his own beliefs about the environment and thinks that we should use a different distribution Q instead of P. Therefore, to evaluate P, he would try to measure how bad using the wrong distribution $P(X \mid e)$ is, while he thinks the true distribution is Q.

Let us assume that a real number can be used for measuring that quantity and denote it as $\mathbb{D}\left(Q_{X|e} \parallel P_{X|e}\right)$. It seems reasonable as P gets closer to Q this measure decreases, and increases when P diverges from Q. Therefore, $\mathbb{D}\left(Q \parallel P\right)$ effectively is a measure of the divergence of P from Q, when Q is the true distribution.

It turns out that if we make some assumptions about the properties of this divergence measure, up to a constant factor, it must be equal to the Kullback–Leibler (KL) divergence. Or in an alternative derivation, by adopting a more minimal set of properties, we can show that if we accept Shannon's entropy as a measure of the uncertainty of a distribution, we should also accept KL divergence as the right way for measuring divergence between distributions:

Theorem 1. *Suppose that $\mathbb{D}\left(Q \parallel P\right)$ is a divergence measure, defined for probability distributions Q and P, which satisfies the following properties:*

Consistency with entropy *If U is the uniform distribution with the same support as Q, then $\mathbb{D}\left(Q \parallel U\right) = \mathbb{H}\left[U\right] - \mathbb{H}\left[Q\right]$, where $\mathbb{H}$ denotes Shannon's entropy.*

Additivity *For any two random variables X, Y*

$$\mathbb{D}\left(Q_{X,Y} \parallel P_{X,Y}\right) = \mathbb{D}\left(Q_X \parallel P_X\right) + \sum_x \mathbb{D}\left(Q_{Y|x} \parallel P_{Y|x}\right) Q(x)\cdot$$

Then, $\mathbb{D}\left(Q \parallel P\right)$ is the Kullback-Leibler divergence.

Proof. Let X be a discrete random variable, and $P(X)$ and $Q(X)$ be two arbitrary distributions for X. We define a new continuous random variable Ω and define

$$p(\Omega = \omega \mid X = x) = \begin{cases} \frac{1}{P(x)} & 0 \leq \omega \leq P(X = x), \\ 0 & \text{otherwise}\cdot \end{cases}$$

Notice that both $P(\Omega \mid X)$ and $P(\Omega, X)$ are uniform distributions, and $\mathbb{H}\left[P_{\Omega|x}\right] = \ln P(x)$, $\mathbb{H}\left[P_{\Omega,X}\right] = 0$.

We define $Q(\Omega \mid x)$ to be an arbitrary distribution that has the same support as $P(\Omega \mid x)$. This ensures that $Q(\Omega, X)$ and $P(\Omega, X)$ have the same support as well. Thus, we have

$$\begin{aligned}\mathbb{D}\left(Q_{\Omega,X} \parallel P_{\Omega,X}\right) &= \mathbb{H}\left[P_{\Omega,X}\right] - \mathbb{H}\left[Q_{\Omega,X}\right] \\ &= \sum_x \int q(\omega, x) \ln q(\omega, x) d\omega \\ &= \sum_x Q(x) \int q(\omega \mid x) \ln q(\omega, x) d\omega,\end{aligned}$$

and

$$\begin{aligned}\mathbb{D}\left(Q_{\Omega|x} \parallel P_{\Omega|x}\right) &= \mathbb{H}\left[P_{\Omega|x}\right] - \mathbb{H}\left[Q_{\Omega|x}\right] \\ &= \ln P(x) + \int q(\omega \mid x) \ln q(\omega \mid x) d\omega \cdot\end{aligned}$$

Now, we use the additivity property to obtain a formula for our divergence measure:

$$\begin{aligned}\mathbb{D}\left(Q_X \parallel P_X\right) &= \mathbb{D}\left(Q_{\Omega,X} \parallel P_{\Omega,X}\right) - \sum_x \mathbb{D}\left(Q_{\Omega|x} \parallel P_{\Omega|x}\right) Q(x) \\ &= \sum_x Q(x) \left(\int q(\omega \mid x) \ln \frac{q(\omega, x)}{q(\omega \mid x)} d\omega - \ln P(x)\right) \\ &= \sum_x Q(x) \left(\ln Q(x) \int q(\omega \mid x) d\omega - \ln P(x)\right) \\ &= \sum_x Q(x) \ln \frac{Q(x)}{P(x)} \cdot \qquad \square\end{aligned}$$

In this theorem, the first property simply states that the entropy of a distribution is related to the divergence of the uniform distribution. The more the divergence is, the less the entropy should be.

The second property tells us that the divergence is an additive quantity. We know that the divergence of $P(X, Y)$ from $Q(X, Y)$ is actually a measure of the generic loss of using $P(X, Y)$ when we believe $Q(X, Y)$ is the true distribution. To calculate this loss, we can assume that X happens before Y. In that way, this loss would include the loss of using $P(X)$, and then, if we observe $X = x$, which can happen with probability $Q(x)$, it would also include the loss of using $P(Y \mid x)$. This property indicates that for combining these individual terms we should use addition.

Returning to our earlier discussion, this implies that to evaluate P our expert should use the KL divergence between P and Q. What if we find a distribution C that has a small KL divergence from any other distribution in our model? In that way, every expert would agree that our distribution is a good choice. Interestingly, this relates to the concept of Bayesian convergence, where the posterior probability converges to the same distribution regardless of the prior distribution.

Even if we cannot find a distribution that has a small divergence from every distribution, still using the distribution that minimizes the maximum divergence seems a reasonable choice.

Definition 3 (Central Distribution)**.** Let $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \dots\}$ be a set of pairs of discrete random variables in a probabilistic model $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$. We define the *divergence measure of* $P \in \mathcal{P}$ *for* $\mathcal{X}$ as follows:

$$\mathbb{D}_{\mathcal{X}}[P] = \max_{Q \in \mathcal{P}} \mathbb{E}_Q \left[\sum_i \mathbb{D}\left(Q_{X_i|e_i} \parallel P_{X_i|e_i}\right) \right], \tag{2}$$

where $\mathbb{E}_Q[\cdot]$ denotes expectation relative to Q and each $\mathbb{D}\left(Q_{X_i|e_i} \parallel P_{X_i|e_i}\right)$ is the Kullback-Leibler divergence between $P(X_i \mid e_i)$ and $Q(X_i \mid e_i)$:

$$\mathbb{D}\left(Q_{X_i|e_i} \parallel P_{X_i|e_i}\right) = \sum_{x_i} Q(x_i \mid e_i) \ln \frac{Q(x_i \mid e_i)}{P(x_i \mid e_i)}.$$

We also define *the central distribution for* $\mathcal{X}$ to be a distribution $C_{\mathcal{X}} \in \mathcal{P}$ which minimizes the divergence measure for $\mathcal{X}$:

$$C_{\mathcal{X}} = \arg\min_{P \in \mathcal{P}} \mathbb{D}_{\mathcal{X}}[P] \cdot \tag{3}$$

Proposition. Using the linearity of expectation, we have

$$\mathbb{D}_{\mathcal{X}}[P] = \max_{Q \in \mathcal{P}} \sum_i \sum_{x_i, e_i} Q(x_i, e_i) \ln \frac{Q(x_i \mid e_i)}{P(x_i \mid e_i)}.$$

In this definition $\mathcal{X}$ plays a crucial role.

Lemma 1. *Assume that* $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ *is a parametric model. Let* $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \dots\}$ *be a set of discrete random variable pairs in* $\mathcal{M}_\Theta$.

For every $P \in \mathcal{P}$ we have

$$\begin{aligned}\mathbb{D}_{\mathcal{X}}[P] &\leq \max_{\theta\in\Theta} \mathbb{E}_{\pi}\left[\sum_i \mathbb{D}\left(\pi_{X_i|e_i,\theta} \parallel P_{X_i|e_i}\right) \middle| \theta\right] \\ &= \max_{\theta\in\Theta} \sum_i \sum_{x_i,e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{P(x_i \mid e_i)}.\end{aligned}$$

Proof. Let R be an arbitrary distribution in $\mathcal{M}_\Theta$, since the Kullback-Leibler divergence is positive, its expected value is also positive, and for every parameter value θ we have

$$\sum_i \sum_{x_i,e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{R(x_i \mid e_i)} \geq 0 \cdot$$

Using this, for every distribution P and every parameter θ, we obtain

$$\sum_i \sum_{x_i,e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \leq \sum_i \sum_{x_i,e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{P(x_i \mid e_i)}.$$

Since this inequality holds for any θ, the maximum of the right-hand side is also greater than or equal to the maximum of the left-hand side w.r.t. θ:

$$\max_{\theta\in\Theta} \sum_i \sum_{x_i,e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \leq \max_{\theta\in\Theta} \mathbb{E}_{\pi}\left[\sum_i \mathbb{D}\left(\pi_{X_i|e_i,\theta} \parallel P_{X_i|e_i}\right) \middle| \theta\right]. \tag{4}$$

$\mathcal{M}_\Theta$ is a parametric model and $R(x_i, e_i) = \int \pi(x_i, e_i \mid \theta) r(\theta) d\theta$, where $r(\theta)$ is a probability distribution. Clearly, for every θ we have

$$\sum_i \sum_{x_i,e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \leq \max_{\theta\in\Theta} \sum_i \sum_{x_i,e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)}.$$

Since r is a distribution and is non-negative, we can multiply this inequality by $r(\theta)$ and integrate. Notice that the right hand side is not a function of θ and it will remain unchanged because $\int r(\theta) d\theta = 1$:

$$\mathbb{E}_R\left[\sum_i \mathbb{D}\left(R_{X_i|e_i} \parallel P_{X_i|e_i}\right)\right] \leq \max_{\theta\in\Theta} \sum_i \sum_{x_i,e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)}.$$

Now from Inequality 4, for *any* distribution P and R in $\mathcal{M}_\Theta$, it follows that

$$\mathbb{E}_R\left[\sum_i \mathbb{D}\left(R_{X_i|e_i} \parallel P_{X_i|e_i}\right)\right] \leq \max_{\theta\in\Theta} \mathbb{E}_{\pi}\left[\sum_i \mathbb{D}\left(\pi_{X_i|e_i,\theta} \parallel P_{X_i|e_i}\right) \middle| \theta\right],$$

which proves the theorem. □

Theorem 2. *Let $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ be a parametric model and $\mathcal{X}$ be a set of pairs of discrete random variables in $\mathcal{M}_\Theta$. If for every value of θ there is a distribution $R \in \mathcal{P}$ such that $R(s) = \pi(s \mid \theta)$ for every $s \in \mathcal{S}$, then for any $P \in \mathcal{P}$ we have:*

$$\mathbb{D}_{\mathcal{X}}[P] = \max_{\theta \in \Theta} \mathbb{E}_\pi \left[\sum_i \mathbb{D}\left(\pi_{X_i|e_i,\theta} \parallel P_{X_i|e_i}\right) \,\middle|\, \theta \right] \cdot \tag{5}$$

Proof. For an arbitrary distribution $P \in \mathcal{P}$, we let

$$\theta^* = \arg\max_{\theta \in \Theta} \mathbb{E}_\pi \left[\sum_i \mathbb{D}\left(\pi_{X_i|e_i,\theta} \parallel P_{X_i|e_i}\right) \,\middle|\, \theta \right] \cdot$$

In $\mathcal{M}_\Theta$ there should be a distribution $R \in \mathcal{P}$ that equals $\pi(\cdot \mid \theta^*)$, and we can write

$$\mathbb{E}_R \left[\sum_i \mathbb{D}\left(R_{X_i|e_i} \parallel P_{X_i|e_i}\right) \right] = \mathbb{E}_\pi \left[\sum_i \mathbb{D}\left(\pi_{X_i|e_i,\theta^*} \parallel P_{X_i|e_i}\right) \,\middle|\, \theta^* \right] \cdot$$

On the other hand, we see

$$\max_{Q \in \mathcal{P}} \mathbb{E}_Q \left[\sum_i \mathbb{D}\left(Q_{X_i|e_i} \parallel P_{X_i|e_i}\right) \right] \geq \mathbb{E}_R \left[\sum_i \mathbb{D}\left(R_{X_i|e_i} \parallel P_{X_i|e_i}\right) \right] \cdot$$

Thus,

$$\mathbb{D}_{\mathcal{X}}[P] \geq \max_{\theta \in \Theta} \mathbb{E}_\pi \left[\sum_i \mathbb{D}\left(\pi_{X_i|e_i,\theta} \parallel P_{X_i|e_i}\right) \,\middle|\, \theta \right] \cdot$$

This inequality and Lemma 1 prove the theorem. □

Definition 4 (Generic Parametric Model)**.** We say a parametric model $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ is *generic*, if for any arbitrary probability distribution r, there is a probability measure $R \in \mathcal{P}$, such that for every $s \in \mathcal{S}$

$$R(s) = \int \pi(s \mid \theta) r(\theta) d\theta \cdot$$

Proposition. A generic parametric model satisfies the conditions of Theorem 2.

Theorem 3 (Minimax theorem [1])**.** *Let $\mathcal{X} \subset \mathbb{R}^n$ and $\mathcal{Y} \subset \mathbb{R}^m$ be compact convex sets. If $f : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ is a continuous function that is concave-convex, i.e.*

- $f(\cdot, y) : \mathcal{X} \to \mathbb{R}$ *is concave for fixed* y*, and*
- $f(x, \cdot) : \mathcal{Y} \to \mathbb{R}$ *is convex for fixed* x*.*

Then we have that

$$\max_{x \in \mathcal{X}} \min_{y \in \mathcal{Y}} f(x, y) = \min_{y \in \mathcal{Y}} \max_{x \in \mathcal{X}} f(x, y) \cdot \tag{6}$$

Theorem 4. *Consider a generic parametric model* $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$*, and let* $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \dots\}$ *be a set of pairs of discrete random variables in* $\mathcal{M}_\Theta$*. Assume that for every* $Q_1, Q_2 \in \mathcal{P}$ *and any real number* $0 \leq \alpha \leq 1$*, there is a distribution* $Q^* \in \mathcal{P}$*, such that for every* i, x_i, e_i *we have*

$$Q^*(x_i \mid e_i) = \alpha Q_1(x_i \mid e_i) + (1 - \alpha) Q_2(x_i \mid e_i) \cdot \tag{7}$$

In this model, the central distribution for $\mathcal{X}$ *can be obtained by:*

$$C_{\mathcal{X}} = \arg\max_{P \in \mathcal{P}} \int \mathbb{E}_\pi \left[\sum_i \mathbb{D}\left(\pi_{X_i|e_i,\theta} \parallel P_{X_i|e_i}\right) \middle| \, \theta \right] p(\theta) d\theta \tag{8}$$

$$= \arg\max_{P \in \mathcal{P}} \int \sum_i \sum_{x_i, e_i} p(x_i, e_i, \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{P(x_i \mid e_i)} d\theta, \tag{9}$$

where $p(x_i, e_i, \theta) = \pi(x_i, e_i \mid \theta) p(\theta)$*.*

Proof. We prove this theorem using the Minimax theorem. For that, we define a functional as follows:

$$f(r, P_{X_i|E_i}) = \int \mathbb{E}_\pi \left[\sum_i \mathbb{D}\left(\pi_{X_i|e_i,\theta} \parallel P_{X_i|e_i}\right) \middle| \, \theta \right] r(\theta) d\theta,$$

where $r(\theta)$ is an arbitrary probability distribution, and $P \in \mathcal{P}$. Since $r(\theta)$ can be any distribution, maximizing with respect to r is like maximizing with respect to θ, and we have

$$C_{\mathcal{X}} = \arg\min_{P \in \mathcal{P}} \max_r f(r, P_{X_i|E_i}) \cdot$$

The domain of r is the set of all probability distributions, which is a compact convex set. Moreover, from Equation 7, it follows that the set of probability distributions for $X_i|E_i$ in $\mathcal{M}_\Theta$ is also a compact convex set. Thus, the domain of $f(r, P_{X_i|E_i})$ is a compact convex set.

For a fixed $P_{X_i|E_i}$, $f(\cdot, P_{X_i|E_i})$ is a linear function, which is both concave and convex. On the other hand, if we let $R(x_i, e_i) = \int \pi(x_i, e_i \mid \theta) r(\theta) d\theta$ for $f(r, \cdot)$ we have

$$f(r, \cdot) = c - \sum_i \sum_{x_i, e_i} R(x_i, e_i) \ln P(x_i \mid e_i) \cdot \tag{10}$$

If we calculate the Hessian matrix of $f(r, \cdot)$, we observe that the matrix is diagonal, and each element of the main diagonal is $\frac{R(x_i|e_i)}{P(x_i|e_i)^2}$. Thus, the hessian matrix of $f(\cdot, P_{X_i|E_i})$ is positive semi-definite and $f(r, \cdot)$ is a convex function. Now, Theorem 3 implies

$$\min_{P \in \mathcal{P}} \max_r f(r, P_{X_i|E_i}) = \max_r \min_{P \in \mathcal{P}} f(r, P_{X_i|E_i}) \cdot \tag{11}$$

From Equation 10 we can see that for an arbitrary fixed r we have

$$\min_{P \in \mathcal{P}} f(r, P_{X_i|E_i}) = \max_{P \in \mathcal{P}} \sum_i \sum_{x_i, e_i} R(x_i, e_i) \ln P(x_i \mid e_i) \cdot$$

We know that the Kullback–Leibler divergence, and therefore its expected value, is non-negative:

$$\mathbb{E}_R \left[\sum_i \mathbb{D} \left(R_{X_i|e_i} \parallel P_{X_i|e_i} \right) \right] \geq 0 \cdot$$

By simple algebra we get

$$\sum_i \sum_{x_i, e_i} R(x_i, e_i) \ln P(x_i \mid e_i) \leq \sum_i \sum_{x_i, e_i} R(x_i, e_i) \ln R(x_i \mid e_i) \cdot$$

Recall that $\mathcal{M}_\Theta$ is a generic parametric model, so $R \in \mathcal{P}$, and we can reach this upper bound by letting $P = R$:

$$\min_{P \in \mathcal{P}} f(r, P_{X_i|E_i}) = \int \sum_i \sum_{x_i, e_i} r(x_i, e_i, \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{R(x_i \mid e_i)} d\theta \cdot$$

By Equation 11 we conclude

$$C_{\mathcal{X}} = \arg\max_{R \in \mathcal{R}} \int \mathbb{E}_\pi \left[\sum_i \mathbb{D} \left(\pi_{X_i|e_i,\theta} \parallel R_{X_i|e_i} \right) \middle| \, \theta \right] r(\theta) d\theta \cdot \qquad \square$$

Theorem 5. *Under the assumptions of Theorem 4, the central distribution for $\mathcal{X}$ satisfies:*

$$\sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{C_{\mathcal{X}}(x_i \mid e_i)} = \lambda,$$

where λ is a constant.

Proof. By Theorem 4, the central distribution for $\mathcal{X}$ maximizes

$$\mathcal{F}\left[p(\theta)\right] = \int \sum_i \sum_{x_i, e_i} p(x_i, e_i, \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{P(x_i \mid e_i)} d\theta,$$

with respect to different choices of the prior distribution $p(\theta)$, subject to the constraint $\int p(\theta) d\theta = 1$.

Applying the Lagrange multiplier technique, the local maximum of $\mathcal{F}$ is a stationary point of the Lagrangian functional:

$$\mathcal{L}\left[p(\theta), \lambda\right] = \int \sum_i \sum_{x_i, e_i} p(x_i, e_i, \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{P(x_i \mid e_i)} d\theta - \lambda \left(\int p(\theta) d\theta - 1 \right) \cdot$$

Let us denote the prior distribution corresponding to the central distribution $C_{\mathcal{X}}$ by $c_{\mathcal{X}}(\theta)$. Let $p(\theta) = c_{\mathcal{X}}(\theta) + \epsilon\eta(\theta)$ be a variation of $c_{\mathcal{X}}(\theta)$, where $\eta(\theta)$ is a continuously differentiable function with $\eta(0) = \eta(1) = 0$, and ϵ is a small constant.

Since $c_{\mathcal{X}}(\theta)$ is an extremizing function for $\mathcal{F}\left[p(\theta)\right]$, the lagrangian $\mathcal{L}\left[\epsilon, \lambda\right]$ has a stationary point where $\epsilon = 0$, and we have

$$\left. \frac{\partial \mathcal{L}\left[\epsilon, \lambda\right]}{\partial \epsilon} \right|_{\epsilon = 0} = 0 \cdot$$

Note that

$$\begin{aligned}
\frac{dp(\theta)}{d\epsilon} &= \eta(\theta), \\
\frac{dP(x_i, e_i)}{d\epsilon} &= \int \pi(x_i, e_i \mid \theta) \eta(\theta) d\theta, \\
\frac{dP(e_i)}{d\epsilon} &= \int \pi(e_i \mid \theta) \eta(\theta) d\theta \cdot
\end{aligned}$$

Using the chain rule, we calculate the partial derivative of $\mathcal{L}\left[\epsilon, \lambda\right]$ relative to ϵ:

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial \epsilon} = & \int \left(\sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{P(x_i \mid e_i)} \right) \eta(\theta) d\theta \\
& - \int \left(\sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \frac{\int \pi(x_i, e_i \mid \theta) \eta(\theta) d\theta}{P(x_i, e_i)} \right) p(\theta) d\theta \\
& + \int \left(\sum_i \sum_{e_i} \pi(e_i \mid \theta) \frac{\int \pi(e_i \mid \theta) \eta(\theta) d\theta}{P(e_i)} \right) p(\theta) d\theta \\
& - \int \lambda \eta(\theta) d\theta
\end{aligned}$$

Since $P(x_i, e_i) = \int \pi(x_i, e_i \mid \theta)p(\theta)d\theta$ and $P(e_i) = \int \pi(e_i \mid \theta)p(\theta)d\theta$ we get

$$\frac{\partial \mathcal{L}}{\partial \epsilon} = \int \left(-\lambda + \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{P(x_i \mid e_i)} \right) \eta(\theta) d\theta\cdot$$

We let $\frac{\partial \mathcal{L}}{\partial \epsilon}\Big|_{\epsilon=0} = 0$. Since $\eta(\theta)$ is an arbitrary function, by applying the fundamental lemma of the calculus of variations and letting $\epsilon = 0$, for every value of θ we have

$$\sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{C_{\mathcal{X}}(x_i \mid e_i)} = \lambda\cdot \qquad \square$$

Theorem 6. *Let $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ be a generic parametric model, and $\mathcal{X} = \{X_1, X_2, X_3, \dots\}$ be a set of random variables in $\mathcal{M}_\Theta$. Let Ω_i, for every i, be a function of the parameter Θ that fully parametrizes X_i, so that for $\omega_i = \Omega_i(\theta)$ we have:*

$$\pi(x_i \mid \theta) = \pi\left(x_i \mid \omega_i\right)\cdot \tag{12}$$

The prior distribution corresponding to the central distribution for $\mathcal{X}$ in $\mathcal{M}_\Theta$ can be characterized by the following fixed point equation:

$$c_{\mathcal{X}}(\theta) \propto \left(\prod_i c_{\mathcal{X}}\theta \mid \omega_i \rho(\beta_i \mid \alpha_i) \right)^{\frac{1}{|\mathcal{X}|}}, \tag{13}$$

where

$$\rho(\beta_i \mid \alpha_i) \propto \exp\left\{ \sum_{x_i} \pi(x_i \mid \omega_i) \ln c_{\mathcal{X}}\omega_i \mid x_i \right\}\cdot$$

Proof. First, we show that the set of distributions of each X_i is a convex set. Let $Q^*(x_i) = \alpha Q_1(x_i) + (1-\alpha)Q_2(x_i)$, for some $Q_1, Q_2 \in \mathcal{P}$, $0 \leq \alpha \leq 1$ and arbitrary i, x_i.

$\mathcal{M}_\Theta$ is a parametric model. Therefore,

$$\begin{aligned} Q^*(x_i) &= \alpha \int \pi(x_i \mid \theta) q_1(\theta) d\theta + (1-\alpha) \int \pi(x_i \mid \theta) q_2(\theta) d\theta \\ &= \int \pi(x_i \mid \theta)(\alpha q_1(\theta) + (1-\alpha) q_2(\theta)) d\theta\cdot \end{aligned}$$

$\alpha q_1(\theta) + (1-\alpha)q_2(\theta)$ is a valid prior distribution and $\mathcal{M}_\Theta$ is a generic parametric model. That implies $Q^* \in \mathcal{P}$.

Thus, by Theorem 5 the central distribution of $\mathcal{X}$ satisfies

$$\sum_i \sum_{x_i} \pi(x_i \mid \theta) \ln \frac{\pi(x_i \mid \theta)}{C_{\mathcal{X}}(x_i)} = \lambda\cdot \tag{14}$$

Moreover, from Equation 15 and Bayes' theorem it follows

$$\begin{aligned}
\sum_i \sum_{x_i} \pi(x_i \mid \theta) \ln \frac{\pi(x_i \mid \theta)}{C_{\mathcal{X}}(x_i)} &= \sum_i \sum_{x_i} \pi(x_i \mid \omega_i) \ln \frac{\pi(x_i \mid \omega_i)}{C_{\mathcal{X}}(x_i)} \\
&= \sum_i \sum_{x_i} \pi(x_i \mid \omega_i) \ln \frac{c_{\mathcal{X}}\omega_i \mid x_i}{c_{\mathcal{X}}(\omega_i)} \\
&= \sum_i \ln \frac{\rho(\beta_i \mid \alpha_i)}{c_{\mathcal{X}}(\omega_i)} \cdot
\end{aligned}$$

Since each Ω_i is a function of Θ, for $\Omega_i(\theta) = \omega_i$ we have $c_{\mathcal{X}}\theta, \omega_i = c_{\mathcal{X}}(\theta)$. Therefore,

$$c_{\mathcal{X}}(\omega_i) = \frac{c_{\mathcal{X}}(\theta)}{c_{\mathcal{X}}\theta \mid \omega_i} \cdot$$

Now we rewrite Equation 17:

$$\begin{aligned}
|\mathcal{X}| \ln c_{\mathcal{X}}(\theta) &= \sum_i \ln \left(c_{\mathcal{X}}\theta \mid \omega_i \rho(\beta_i \mid \alpha_i) \right) - \lambda \\
c_{\mathcal{X}}(\theta) &= \left(e^{-\lambda} \prod_i c_{\mathcal{X}}\theta \mid \omega_i \rho(\beta_i \mid \alpha_i) \right)^{\frac{1}{|\mathcal{X}|}}
\end{aligned}$$

□

Theorem 7. *Let $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ be a generic parametric model, and $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \dots\}$ be a set of random variables in $\mathcal{M}_\Theta$. Let Ω_i, for every i, be a function of the parameter Θ that fully parametrizes X_i, so that for $\omega_i = \Omega_i(\theta)$ we have:*

$$\pi(x_i \mid \theta) = \pi\left(x_i \mid \omega_i\right) \cdot \tag{15}$$

The prior distribution corresponding to the central distribution for $\mathcal{X}$ in $\mathcal{M}_\Theta$ can be characterized by the following fixed point equation:

$$c_{\mathcal{X}}(\beta \mid \alpha) \propto \left(\prod_i \frac{c_{\mathcal{X}}(\alpha, \beta \mid \alpha_i, \beta_i) \rho(\beta_i \mid \alpha_i)}{c_{\mathcal{X}}(\alpha \mid \alpha_i)} \right)^{\frac{1}{|\mathcal{X}|}}, \tag{16}$$

where

$$\rho(\beta_i \mid \alpha_i) \propto \exp\left\{ \sum_{x_i, e_i} \pi(x_i, e_i \mid \alpha_i, \beta_i) \ln c_{\mathcal{X}}(\beta_i \mid x_i, e_i) \right\} \cdot$$

Proof. First, we show that the set of distributions of each X_i is a convex set. Let $Q^*(x_i \mid e_i) = \alpha Q_1(x_i \mid e_i) + (1 - \alpha) Q_2(x_i \mid e_i)$, for some $Q_1, Q_2 \in \mathcal{P}$, $0 \le \alpha \le 1$ and arbitrary i, x_i, e_i.

$\mathcal{M}_\Theta$ is a parametric model. Therefore,

$$\begin{aligned} Q^*(x_i \mid e_i) &= \alpha \int \pi(x_i \mid e_i, \beta) q_1(\beta \mid e_i) d\beta + (1-\alpha) \int \pi(x_i \mid \theta) q_2(\theta) d\theta \\ &= \int \pi(x_i \mid \theta)(\alpha q_1(\theta) + (1-\alpha) q_2(\theta)) d\theta \cdot \end{aligned}$$

$\alpha q_1(\theta) + (1-\alpha)q_2(\theta)$ is a valid prior distribution and $\mathcal{M}_\Theta$ is a generic parametric model. That implies $Q^* \in \mathcal{P}$.

Thus, by Theorem 5 the central distribution of $\mathcal{X}$ satisfies

$$\sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{C_\mathcal{X}(x_i \mid e_i)} = \lambda \cdot \tag{17}$$

Moreover, from Equation 15 and Bayes' theorem it follows

$$\begin{aligned} \lambda &= \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{C_\mathcal{X}(x_i \mid e_i)} \\ &= \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \alpha_i, \beta_i) \ln \frac{\pi(x_i \mid e_i, \beta_i)}{C_\mathcal{X}(x_i \mid e_i)} \\ &= \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \alpha_i, \beta_i) \ln \frac{c_\mathcal{X}(\beta_i \mid x_i, e_i)}{c_\mathcal{X}(\beta_i \mid e_i)} \\ &= \sum_i \ln \frac{\rho(\beta_i \mid \alpha_i)}{c_\mathcal{X}(\beta_i \mid \alpha_i)} \cdot \end{aligned}$$

Since each Ω_i is a function of Θ, for $\Omega_i(\theta) = \omega_i$ we have $c_\mathcal{X}\theta, \alpha_i, \beta_i = c_\mathcal{X}(\theta)$ $c_\mathcal{X}\alpha, \alpha_i = c_\mathcal{X}\alpha$. Therefore,

$$c_\mathcal{X}(\beta_i \mid \alpha_i) = \frac{c_\mathcal{X}(\alpha \mid \alpha_i)}{c_\mathcal{X}(\alpha, \beta \mid \alpha_i, \beta_i)} c_\mathcal{X}(\beta \mid \alpha) \cdot$$

Now we rewrite Equation 17:

$$|\mathcal{X}| \ln c_\mathcal{X}(\beta \mid \alpha) = -\lambda + \sum_i \ln \frac{c_\mathcal{X}(\alpha, \beta \mid \alpha_i, \beta_i) \rho(\beta_i \mid \alpha_i)}{c_\mathcal{X}(\alpha \mid \alpha_i)}$$

Then:

$$\begin{aligned}
c_{\mathcal{X}}(\beta \mid \alpha) &= \left(e^{-\lambda} \prod_i \frac{c_{\mathcal{X}}(\alpha, \beta \mid \alpha_i, \beta_i) \rho(\beta_i \mid \alpha_i)}{c_{\mathcal{X}}(\alpha \mid \alpha_i)} \right)^{\frac{1}{|\mathcal{X}|}} \\
&= \left(e^{-\lambda} \prod_i \frac{c_{\mathcal{X}}(\beta \mid \alpha) \rho(\beta_i \mid \alpha_i)}{c_{\mathcal{X}}(\beta_i \mid \alpha_i)} \right)^{\frac{1}{|\mathcal{X}|}} \\
&= c_{\mathcal{X}}(\beta \mid \alpha) \left(\prod_i \frac{\rho(\beta_i \mid \alpha_i)}{c_{\mathcal{X}}(\beta_i \mid \alpha_i)} \right)^{\frac{1}{|\mathcal{X}|}} \\
&= c_{\mathcal{X}}(\beta \mid \alpha) \left(e^{-\lambda} \prod_i \frac{\rho(\alpha_i, \beta_i) c_{\mathcal{X}}(\alpha_i)}{c_{\mathcal{X}}(\alpha_i, \beta_i) \rho(\alpha_i)} \right)^{\frac{1}{|\mathcal{X}|}} \qquad \square
\end{aligned}$$

References

[1] Ding-Zhu Du. *Minimax and Its Applications*, pages 339–367. Springer US, Boston, MA, 1995.